

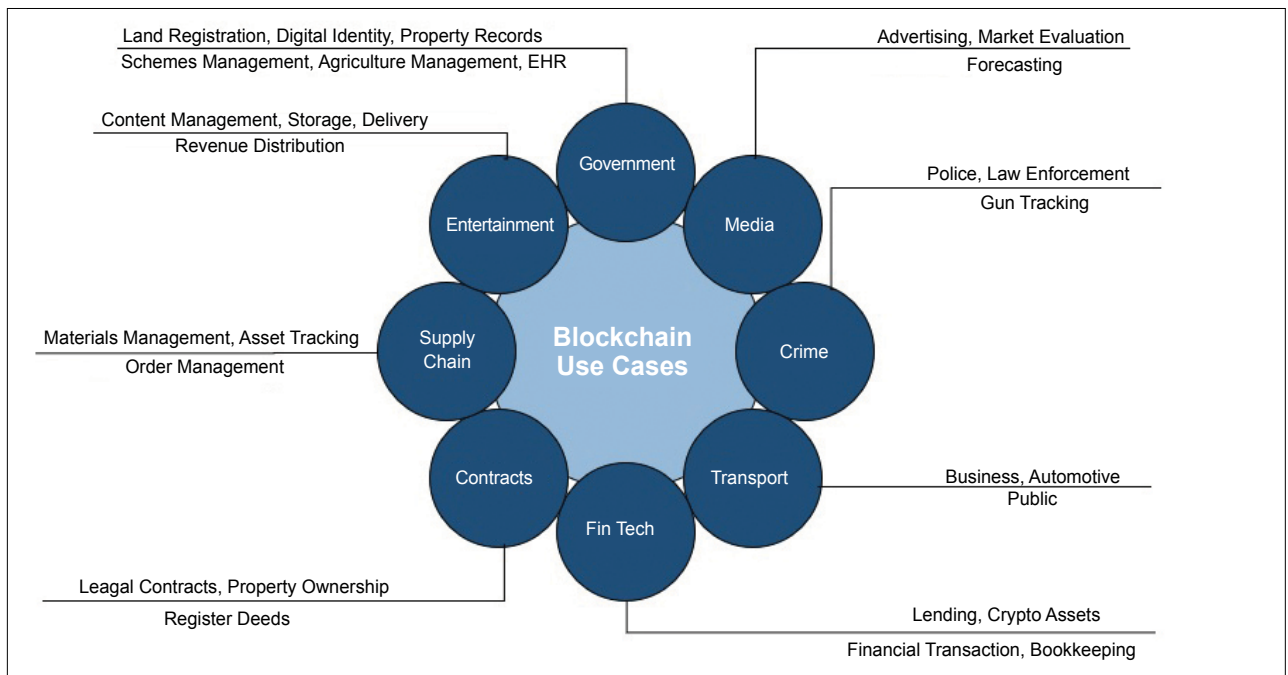


Figure 3: Blockchain use cases

verify the owner of a house quickly and easily.

- **Revenue:** Blockchain enables seamless cross-referencing of documentation across multiple government entities. All documentation is automatically verified by the relevant entities. The inherent features of blockchain (digital identity) make it impossible to make changes to the ledger.
- **Managing entity:** Business unit incorporation and governance is possible with the use of digital signatures and programmable business rules. It also enables approvals and validations of the entity and its application on the blockchain. The distributed ledger technology (DLT) allows multiple parties to digitally sign for authorisation and approve a transaction and documents.
- **Grants:** An integrated blockchain solution enables seamless customer onboarding with automated disbursements of loans and grants. Transactions on the ledger are auditable in real-time providing greater transparency for regulators. Highly secured digital signatures make sure transactions get executed only when the relevant entities provide their signature.
- **Agriculture:** Integrated agriculture management system (IAMS) with a blockchain infrastructure ensures immutability of data and a way to trace historical agricultural data, as agricultural products move from production sources to the consumer. IAMS with a blockchain infrastructure helps diversify current agricultural management practices in a way that engages the public through ownership of the agricultural production process. Other possible applications include the use of

blockchain technology to record and manage agricultural land records as well as agriculture insurance.

- **Healthcare:** Digitisation of health records is a significant task in the public health sector, which is huge, complex and associated with ethical issues. Medical records are scattered and erroneous, with inconsistent data handling processes. Sometimes, hospitals and clinics are forced to work with incorrect or incomplete patient records. Blockchain technology can help create a secure and flexible ecosystem for exchanging a patient's electronic health records. This technology can also make the space more transparent by creating a basis for critical drugs, blood, organs, etc. In addition, by putting all medical licences on a blockchain, fraudulent medical practitioners can be prevented from practising, and chemists can be prevented from selling spurious drugs.
- **Education:** Student records, faculty records, educational certificates, etc, are key assets in the education domain and are shared with multiple stakeholders; so it is imperative to ensure that they are trustworthy. All these records can be maintained with the application of blockchain technology. Blockchain can also simplify certificate attestation and verification.

Media

Media is a multi-billion dollar industry that is evolving every year. However, it is currently not consumer- or creator-friendly due to intermediaries taking advantage and making tons of revenue in the process. With blockchain, the